

Steel Beam Estimation – Ground Truth Benchmark Summary

Version 10.0.0 | All Six Drawing Sets — Known (QA.2B.1) + Unseen (QA.3.0)

Benchmark Results

This report consolidates ground-truth accuracy across all six drawing sets. First–Third Set results are taken from Version 9 QA.2B.1 regenerated production outputs (known / development drawing sets). Fourth–Sixth Set results are taken from Version 10 QA.3.0 unseen generalization benchmark production outputs. For every set, model Estimation_Output.xlsx was compared against the estimator bar bending schedule (ground truth only). Estimator Excel was never used during DXF production.

Accuracy Summary vs Version 9.6.1 (First–Third Baseline)

The table below compares the Version 9.6.1 three-set (First–Third) baseline against the combined six-set accuracy (First–Sixth).

Metric	V9.6.1 First–Third	V10 All Six Sets	Delta
Beam Detection	93.92%	85.55%	-8.37%
Bar Detection	68.52%	50.99%	-17.53%
Bar Matching Accuracy	27.41%	33.98%	+6.57%
Overall Accuracy	63.29%	56.84%	-6.45

Known vs Unseen Subgroup Accuracy

First–Third are known development sets (QA.2B.1). Fourth–Sixth are unseen generalization sets (QA.3.0).

Metric	First–Third (Known)	Fourth–Sixth (Unseen)	All Six Sets
Beam Detection	93.92%	82.95%	85.55%
Bar Detection	68.52%	46.04%	50.99%
Bar Matching	27.41%	36.74%	33.98%
Overall Accuracy	63.29%	55.24	56.84%

Drawing-wise Performance

Drawing Set	Beam Detection	Bar Detection	Bar Accuracy	Overall
First Set	18 / 18 (100.0%)	81.18%	34.78%	71.9%
Second Set	64 / 67 (95.5%)	71.76%	35.74%	67.7%

Third Set	57 / 63 (90.5%)	63.52%	17.91%	57.3%
Fourth Set	112 / 143 (78.3%)	39.89%	37.33%	51.8%
Fifth Set	143 / 187 (76.5%)	40.25%	34.30%	50.4%
Sixth Set	139 / 145 (95.9%)	61.45%	38.87%	65.4%
All Six Sets	533 / 623 (85.55%)	50.99%	33.98%	56.7%

Across all six sets, beam detection remains relatively strong, but drops on unseen Fourth/Fifth projects. Steel accuracy is high on First and Sixth sets and weaker on Fourth/Fifth. Reinforcement bar detection and matching remain the primary limiters of overall estimation accuracy, especially under unseen generalization conditions.

Overall Performance Summary

Beam Detection

- Ground Truth Beams: **623**
- Correctly Detected Beams: **533**
- **Beam Detection Accuracy: 85.55%**

Reinforcement Bar Interpretation

- Ground Truth Bars: **4254**
- Detected Bars: **2169**
- Correctly Matched Bars: **737**
- Missing Bars: **2085**
- **Bar Detection Accuracy: 50.99%**
- **Bar Matching Accuracy: 33.98%**

Overall Model Accuracy

- Overall Ground Truth Benchmark Accuracy (All Six Sets): 56.7%

Key Error Categories

The benchmark automatically classified discrepancies between regenerated model outputs and estimator ground truth across all six drawing sets (combined QA.2B.1 + QA.3.0 error tallies).

Error Category	Count
Missing Bars	2085
Wrong Quantity	660
Extra Bars	447
Wrong Diameter	526
Wrong Role Classification	167
Beam Missing	90
Beam Mismatch	14
Steel Difference	5

Most of the errors remain associated with **reinforcement interpretation**, particularly **missing bars, wrong quantity, extra bars, and diameter resolution**. Total classified errors (all six sets): 3994.

Benchmark Scope and Engineering Context

This six-set report spans the Version 9 known-set regeneration programme and the Version 10 unseen generalization benchmark:

- First–Third Set: QA.2B.1 production regeneration (known development drawings)
- Fourth–Sixth Set: QA.3.0 unseen generalization benchmark
- DXF-only production; estimator Excel used only for post-run GT comparison
- Beam Ownership / Annotation Graph / Adaptive Extent / Shared Scope (Track 1)
- Stirrup recovery and geometric evidence pipeline
- P2.1–P2.4 leader-chain / bar-failure diagnostic programme (no formula change in this report)